

Prior Predictive Checks: Grammaticality Judgment Example

Bayesian Mixed Effects Models with brms for Linguists

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1 Prior Predictive Checks for Binary Data

This document demonstrates how to validate priors for a Bayesian binary model (grammaticality judgments) before fitting to data.

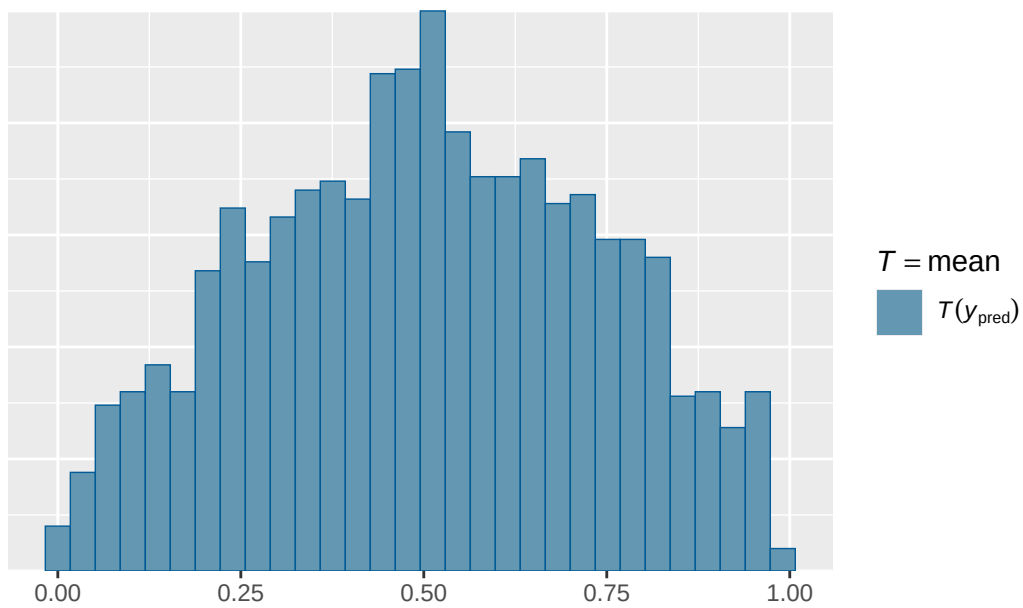
1.1 Setup

1.2 Fitting Prior Only Model

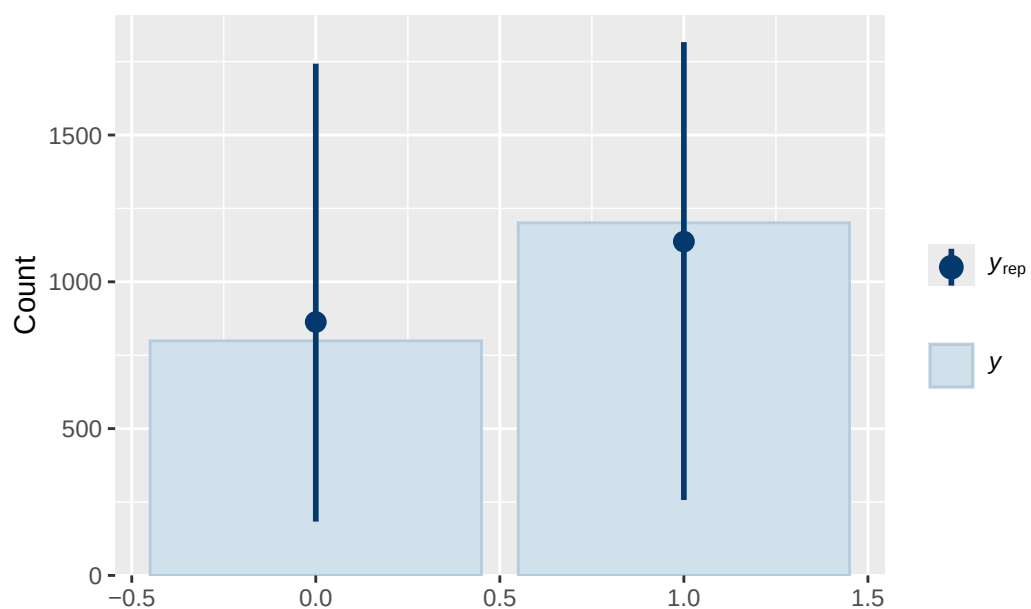
1.3 Prior Predictive Checks

1.3.1 Visual Checks

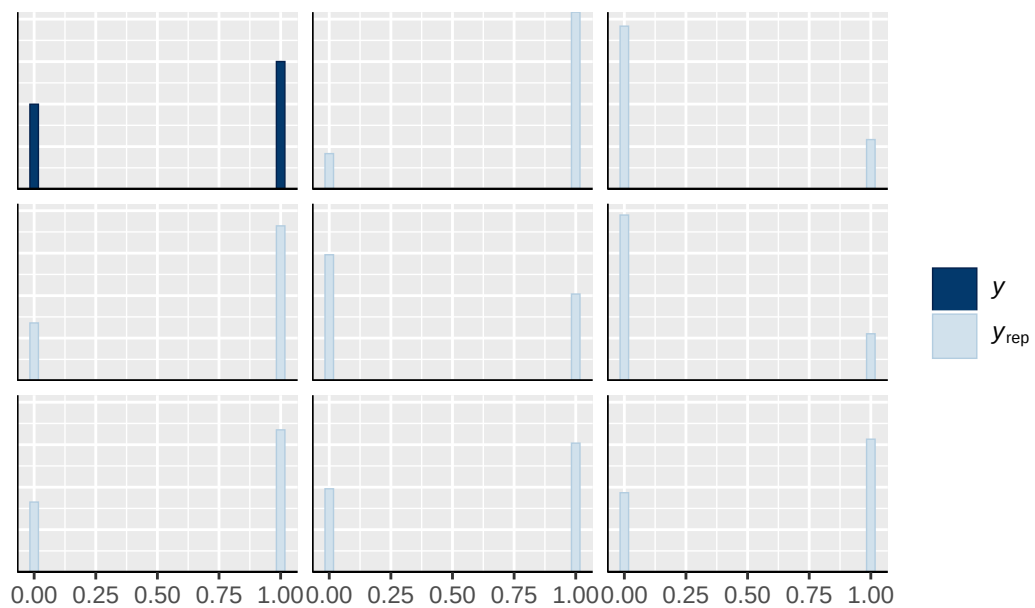
Prior Predictions vs Observed Accuracy



Observed vs Predicted Counts

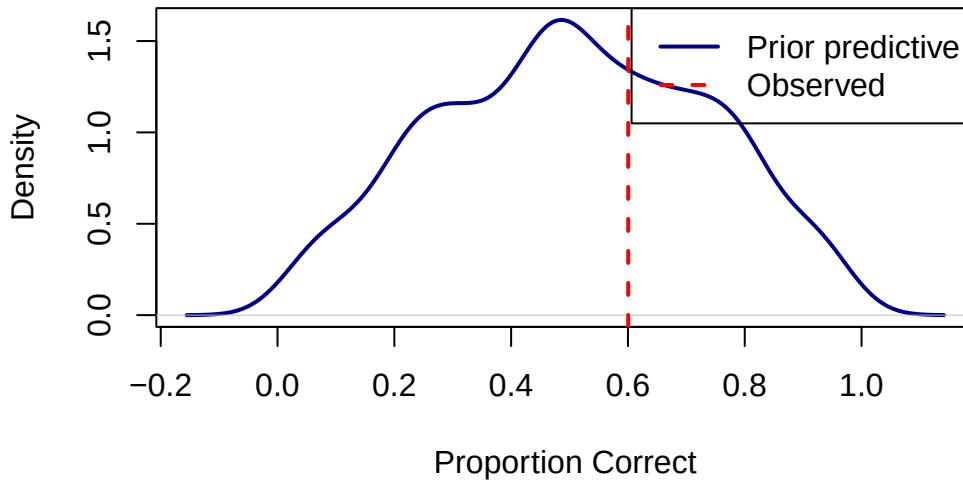


Histograms: Simulated Data from Prior



1.3.2 Prior Predictive Accuracy Distribution

Prior Predictive Distribution of Accuracy



1.4 Prior Distributions

Following Kurz's approach, we extract prior samples using `as_draws_df()` from the posterior package.

1.4.1 Intercept Prior

Intercept prior (log-odds scale):

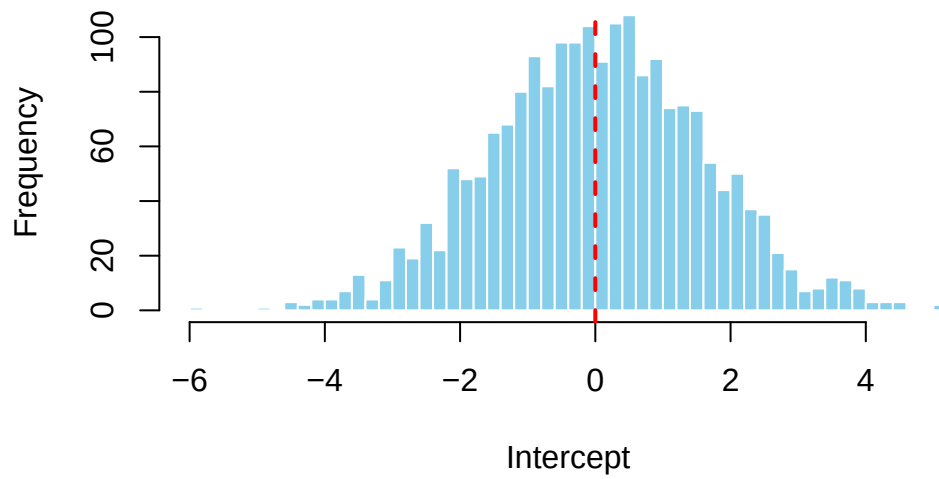
2.5%	50%	97.5%
-2.98728595	0.04194952	3.19327008

Intercept prior (probability scale):

2.5%	50%	97.5%
0.04800357	0.51048584	0.96058023

Prior expects 5 - 96 % baseline accuracy (median 51 %)

Prior for Intercept (log-odds scale)



1.4.2 Effect Size Prior

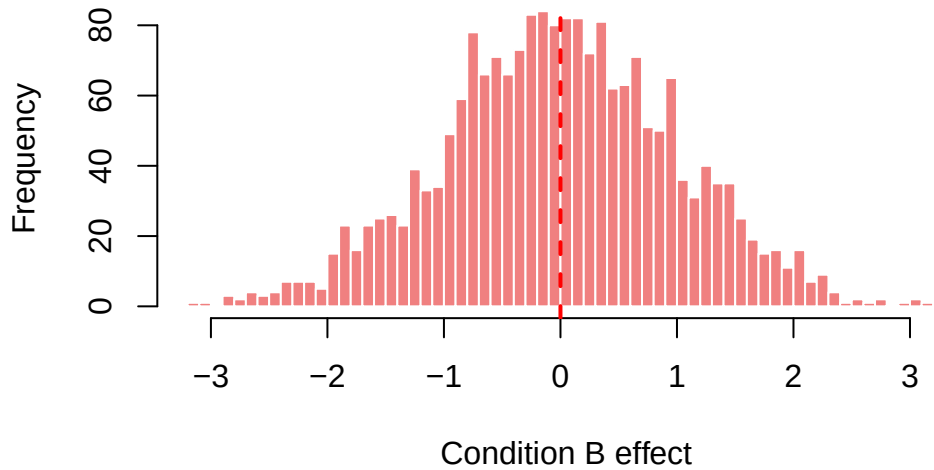
Condition effect prior (log-odds scale):

2.5%	50%	97.5%
-1.93961957	-0.01510226	1.96317214

Condition effect (odds multiplier):

2.5%	50%	97.5%
0.1437586	0.9850112	7.1218829

Prior for Condition Effect (log-odds scale)



1.5 Random Effects Distributions

For prior predictive checks, we examine the **implied distribution** of subject-specific parameters by extracting the hyperprior SDs and simulating random effects.

1.5.1 Subject Random Intercepts

Subject random intercept SD prior:

2.5%	50%	97.5%
0.02900055	0.68343780	3.86970012

Implied subject random intercepts (log-odds scale):

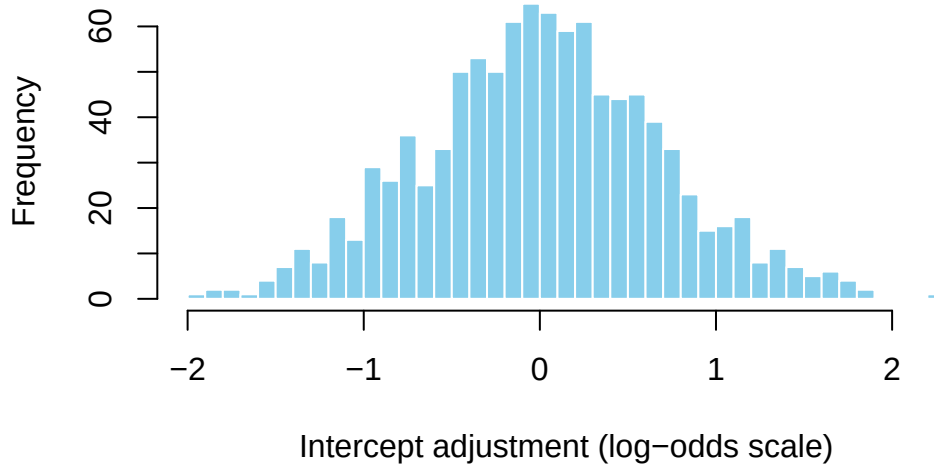
2.5%	50%	97.5%
-1.326931618	0.006294215	1.392768861

Implied subject-specific accuracy (probability scale):

2.5%	50%	97.5%
0.2167034	0.5120586	0.8076354

Prior implies subject accuracy ranges from 22 % to 81 %

Prior-implied Subject Random Intercepts



1.5.2 Subject Random Slopes

Subject random slope SD prior:

2.5%	50%	97.5%
0.02717447	0.68593886	3.63193775

Implied subject random slopes (log-odds scale):

2.5%	50%	97.5%
-1.36609138	0.03762538	1.30733868

Implied subject-specific effects (odds multipliers):

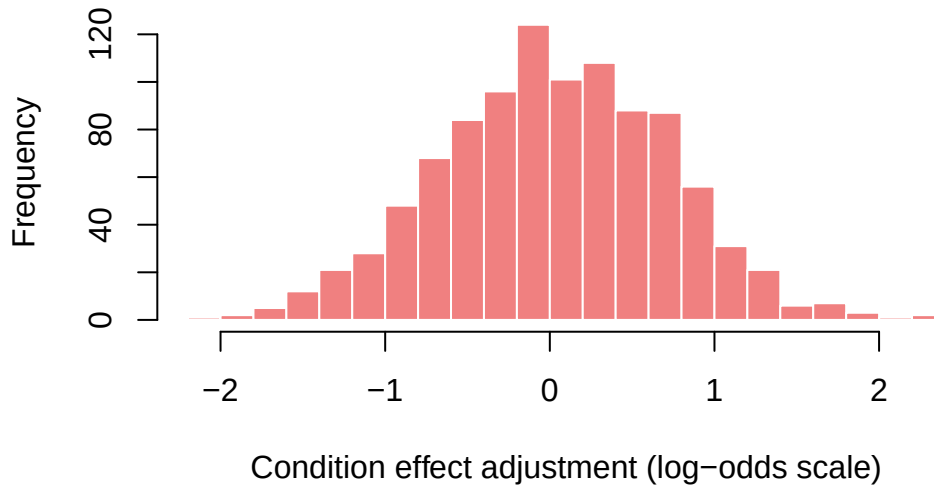
2.5%	50%	97.5%
0.2551021	1.0383422	3.6963235

Weak effect subjects (2.5%): $0.26 \times$ odds multiplier

Average effect subjects (50%): $1.04 \times$ odds multiplier

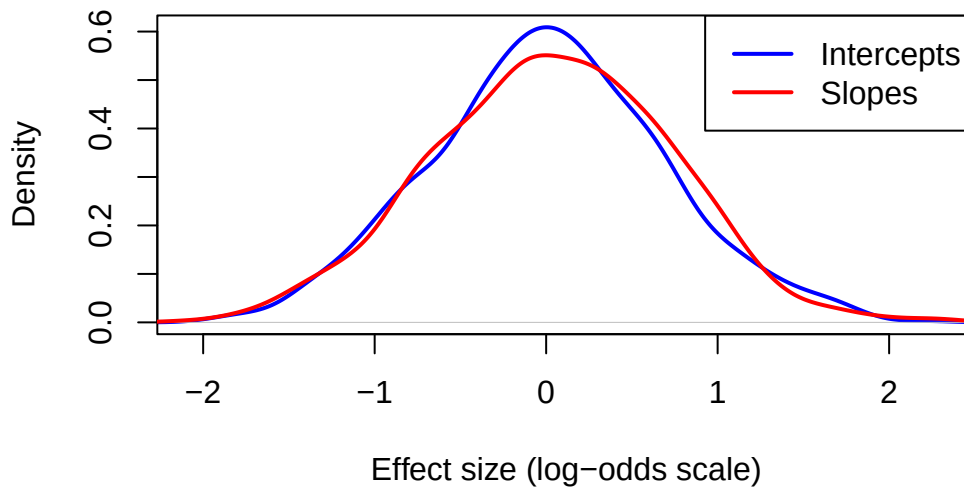
Strong effect subjects (97.5%): $3.7 \times$ odds multiplier

Prior-implied Subject Random Slopes



1.5.3 Random Effects Comparison

Prior-implied Random Effects Distributions



1.6 Interpretation

1.6.1 Good Signs (Prior is Reasonable)

- Prior generates ~50% baseline accuracy (Intercept = 0)
- Condition effect varies but is typically moderate
- Between-subject accuracy ranges 30-70% or similar
- No impossible values (all between 0 and 1)

1.6.2 Problems to Watch For

- Mean accuracy » 90%: intercept prior too high
- All subjects $50\% \pm 1\%$: intercept SD too small
- Very weak prior predictions: slopes prior too small

1.7 Summary

Before fitting your model to actual data, always validate that your priors: 1. Generate reasonable predictions for your domain 2. Allow sufficient flexibility for the data to inform the posterior 3. Respect constraints (e.g., 0-1 for probabilities) 4. Account for variation across subjects/items

Adjust priors as needed and rerun these checks.